INSIDE THE EARTH

Interpreting earthquake patterns through geography, time, depth and magnitude

DSC5002 Project Team | City University of Hong Kong | USGS catalogue, 2000–2025

46,463 EARTHQUAKES

M ≥ 5 | 26 years | 5 questions

One frozen UTC catalogue

ABSTRACT Geographic references show strong boundary concentration, while regional depth tails and event windows expose structure hidden by pooled summaries. Time contrasts depend on the magnitude threshold and coincide with shifts in catalogue composition. We connect maps, exact comparisons and sensitivity checks to distinguish descriptive evidence from unsupported claims about causes or future hazard.

01 STUDY DESIGN

What can one catalogue show?

Earthquakes vary in location, depth, magnitude and timing. A map identifies clusters, but does not separate geographic concentration, large event sequences and changes in reporting. We ask where recorded activity concentrates, how its time pattern changes, and which regional structures disappear when observations are pooled.

Table 1. Shared analytical scope

Element	Definition
Window	2000–2025 UTC
Selection	Earthquake; preferred M ≥ 5
Observations	46,463 unique event IDs
Regions	6 ordered boxes + Other
Distance	PB2002 spherical arcs
Depth bands	Shallow <70; intermediate 70–<300; deep ≥300 km

From question to comparison

Space: compare boundary distances with equal-area and latitude-controlled reference points. **Time:** compare equal 13-year periods and inspect reporting changes. **Depth:** contrast regional distributions and local sections. **Magnitude and cases:** separate frequency, a relative proxy and time-window concentration.

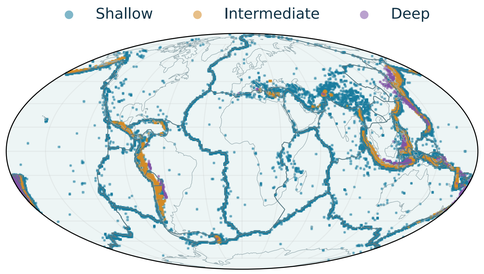


Fig. 1. All 46,463 epicentres; equal-area Mollweide projection. Colours denote depth bands; grey lines show PB2002 boundaries. Overlapping points are not an area-normalized density.

Data checks and statistical choices

Coordinates, magnitude and depth are present for every record. Event IDs are deduplicated and non-earthquake types excluded. Four negative reported depths are retained. A complete download is not proof of complete detection; preferred magnitudes are not a homogeneous measurement scale.

Medians summarize typical values, tail shares retain rare deep events, and rank correlations describe monotonic association. Counts use all eligible records. We avoid independent-event p-values because clustered sequences do not supply independent observations.

Why the event-type filter matters

An all-type companion extract has 46,523 records: 55 volcanic eruptions, four nuclear explosions and one landslide must be excluded. The remaining 46,463 earthquake IDs and principal measurements match the analysis catalogue. Treating the 60-record difference as a catalogue revision would misstate the sample.

Reading the evidence

Each column moves from a defined comparison to a result and its interpretation. Unequal region areas, selected routes, catalogue composition and window definitions remain visible rather than being absorbed into a single global conclusion.

02 SPACE & REFERENCE

Concentrated relative to what?

Boundary proximity needs a geographic denominator. We compare event distances with equal-area points over the whole globe, including oceans. A second reference preserves the observed latitude distribution while varying longitude.

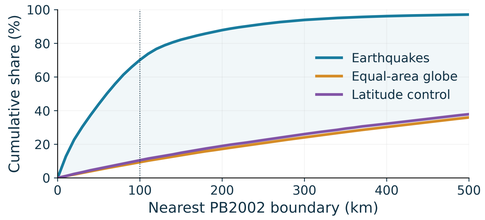


Fig. 2. Cumulative shares under identical PB2002 minor great-circle arc geometry, R = 6,371.0088 km. The reference curves answer a geographic comparison, not a seismicity null hypothesis.

Table 2. Within-distance shares (%)

Radius (km)	Events	Equal area	Latitude control
50	43.92	4.96	5.54
100	70.15	9.42	10.50
200	87.90	17.27	18.99
500	97.13	35.98	37.88

At 100 km, **70.15%** of events versus **9.42%** of equal-area points gives an observed-to-area share ratio of **7.45**. Latitude control gives **10.50%**. At 500 km the ratio narrows to 2.70: the threshold changes the contrast.

Does every region behave alike?

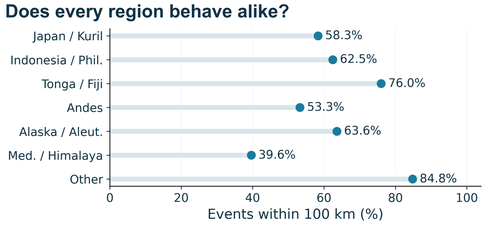


Fig. 3. Within-region percentages inside 100 km. Each region uses its own earthquake denominator; these are not area-normalized hazard rates.

Tonga/Fiji has a 76.0% nearby share, versus 39.6% in Mediterranean/Himalaya. The high 84.8% share in Other belongs to a heterogeneous residual group; it is not one coherent tectonic region.

Table 3. Magnitude-threshold check

Cutoff	N	≤100 km (%)	Median (km)
M ≥ 5	46,463	70.15	59.7
M ≥ 6	3,874	65.93	68.5
M ≥ 7	387	63.57	79.4

The within-100-km share falls to **63.57%** at M ≥ 7. Broad boundary concentration does not imply the same proximity distribution at every magnitude threshold.

Reference sensitivity

Four 16,384-point equal-area designs give 9.38–9.53% within 100 km. This is numerical sensitivity, not a confidence interval. The 7.45 ratio is descriptive; simplified PB2002 proximity does not quantify fault uncertainty or future risk.

03 TIME & COMPARABILITY

A fluctuating recorded history

Annual M ≥ 5 counts peak at **2,701 in 2011**. Equal-period means, a whole-period slope and regional changes answer different questions, so no single summary is treated as the trend.

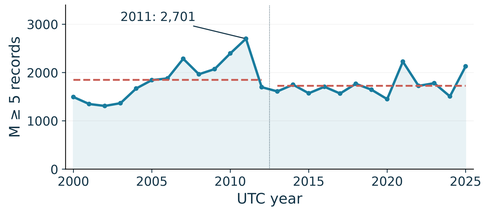


Fig. 4. Annual counts and 13-year means (dashed). Comparisons account for 4,749 early-period days and 4,748 late-period days.

Table 4. Catalogue comparability

Indicator	2000–2012	2013–2025
M ≥ 5 records	24,030	22,433
Annual mean	1848.5	1725.6
Mw-family share	73.45%	55.05%
Horizontal-error availability	0.28%	86.80%

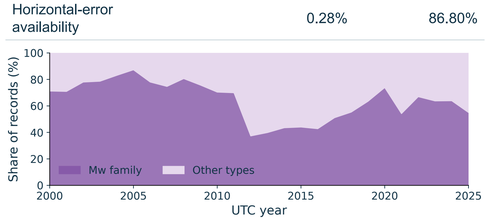


Fig. 5. Annual Mw-family composition exposes changes concealed by the two period averages. Selecting Mw alone does not make measurement homogeneous.

Later-period rates are **6.63% lower at M ≥ 5** and **15.42% lower at M ≥ 6**, yet the full-period M5 Sen slope is positive. Changing magnitude composition and error reporting limit interpretation as changing physical activity. Missing error does not mean zero error.

The direction depends on region

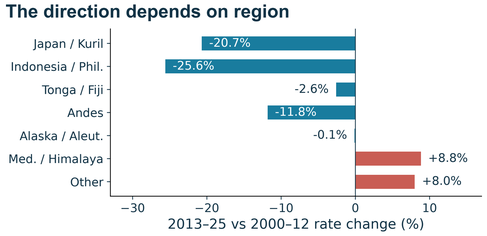


Fig. 6. Exposure-adjusted M5 rate changes, 2013–2025 versus 2000–2012. Shared periods support within-region comparison; unequal box areas prevent direct hazard ranking.

Indonesia/Philippines falls by **25.58%**; Mediterranean/Himalaya rises by **8.82%**. Aggregation conceals that contrast. Selecting only Mw-family records also changes the sample and does not homogenize the catalogue.

Large-event window check

Removing the union of 30-day / 500-km windows around 28 M ≥ 8 anchors removes 3,090 records and changes the M5 contrast to −3.37%. Anchors are included in this removal. This is sensitivity analysis, not declustering.

04 DEPTH & ASSOCIATIONS

Similar medians, different tails

Globally **83.20%** of events are shallow. Japan/Kuril and Tonga/Fiji both have median depth 33 km, but deep-event shares differ by more than twofold. A centre alone hides that tail contrast.

Table 5. Regional depth summaries

Region	N	Median (km)	Deep (%)
Global	46,463	23.7	3.71
Japan / Kuril	4,954	33.0	5.01
Tonga / Fiji	8,904	33.0	11.76
Andes	3,769	35.0	2.60

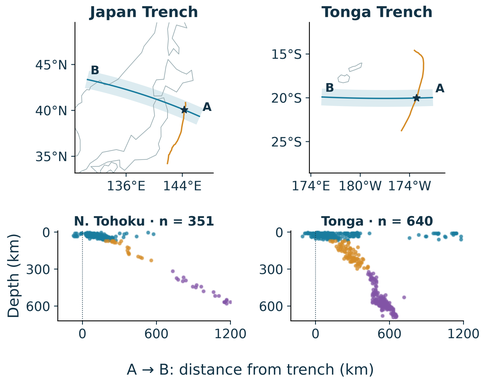


Fig. 7. Gold trench; blue route and ±100 km swath; star at 0 km. A → B spans −200 to +1,200 km. Depth and distance use equal kilometre scales. Routes were chosen after exploration.

Selected-section deep counts are **21/351** in northern Tohoku and **186/640** in central Tonga. These local swaths are subsets, so their shares differ from Table 5. Tonga's selected 29.1% deep share versus 11.76% regionally illustrates why a section cannot represent its entire parent region.

Does the pooled association hold?

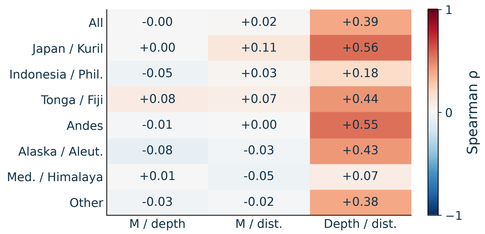


Fig. 8. Spearman ρ for all records and each region. Common −1 to +1 scale; descriptive associations, without independent-event significance tests.

Depth–boundary-distance ρ is **+0.39 globally** and **+0.56 in Japan/Kuril**, but only +0.07 in Mediterranean/Himalaya. Magnitude associations are weak overall. Pooled coefficients do not imply identical regional relationships.

Geological interpretation

Depth tails and local sections reveal structure consistent with regional differences. Selected routes, simplified boundaries and observational associations do not independently identify the physical mechanism or prove causality.

05 MAGNITUDE & SEQUENCES

Frequency and scale are different

An event-count distribution describes how often magnitudes occur. A summed magnitude-based proxy gives much greater weight to rare, large events; the two shares should be compared explicitly.

Table 6. Counts and relative proxy

M band	N	Events (%)	Proxy (%)
5–<6	42,589	91.66	2.00
6–<7	3,487	7.50	5.52
7–<8	359	0.77	21.27
≥8	28	0.06	71.21

Using $R(M)=10^{1.5(M-5)}$, one magnitude unit multiplies the proxy by 31.6. Only **28 events at M ≥ 8** supply **71.21%** of its sum. Mixed preferred scales make this illustrative, not measured joules, damage or risk.

Tail counts and completeness

Inclusive N(M ≥ m) is **3,874 at M ≥ 6**, **387 at M ≥ 7** and **28 at M ≥ 8**. The M5 extraction threshold does not establish completeness, so no predictive b-value is fitted.

Counts first; timing second

Cases use radius ≤500 km and UTC windows [−30,0) and [0,30) days relative to the exact anchor time. Anchors are excluded. All windows are fully observed.

Table 7. Fixed-window case counts

Case	Before	After	First 3d (% after)
Sumatra 2004	2	143	60.1
Maule 2010	1	302	60.9
Tohoku 2011	33	563	70.9

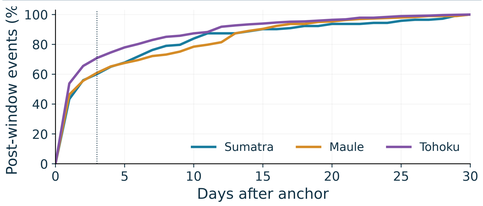


Fig. 9. Each curve divides by its own 30-day post-window count. Day d includes [0,d). Normalization separates the timing pattern from unequal totals.

The first three days contain **60.1%, 60.9% and 70.9%** of the post-window records. Early concentration coexists with a longer tail; changing duration changes the total being summarized.

Table 8. Duration sensitivity at 500 km

Case	7 days	30 days	90 days
Sumatra 2004	109	143	179
Maule 2010	218	302	331
Tohoku 2011	467	563	647

Post-window counts use the same radius and anchor exclusion. These are nested samples, not independent replicates. The full supplement also varies radius from 250 to 1,000 km.

What the cases establish

The selected sequences show concentrated recorded activity after major events. Time-and-distance membership alone does not establish causal aftershock status or a universal sequence law.

CONCLUSION Geographic references establish a strong descriptive concentration near mapped boundaries. Depth tails and regional associations reveal structure that global summaries conceal; event curves separate timing from totals. Time contrasts require catalogue-comparability checks, and window choices materially change case counts. These are reproducible comparisons of recorded patterns, not forecasts of future hazard.

[1] USGS FDSN · [2] Bird (2003), PB2002; GeoJSON (ODC-By) · [3] Natural Earth 110m (public domain) · [4] USGS magnitude–energy guidance. Full methods, all figures and sensitivity results: project repository.